

Road Segmentation

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Abstract—Segmenting roads in satellite images is essential for urban planning, autonomous navigation, and infrastructure monitoring. This study investigates the use of machine learning to classify pixels as roads or background using data from Google Maps, evaluated using the F1-score. Starting with traditional models like logistic regression and Multi-Layer Perceptrons (MLPs) as baselines, the approach progressed to Convolutional Neural Networks (CNNs), which leveraged spatial patterns to improve accuracy. The final and most effective step was the use of U-Net architectures, which combined detailed local features with broader contextual information, achieving an F1-score of 0.82. These findings demonstrate the potential of deep learning for automated road detection and its applicability to real-world scenarios.

I. INTRODUCTION

Accurate road segmentation is essential for urban development, transportation planning, and disaster response. With cities expanding, satellite imagery offers a valuable resource for analyzing large areas efficiently.

Manual analysis of satellite images is labor-intensive and error-prone, but machine learning provides a powerful alternative, enabling automated road detection with greater accuracy. Using a dataset of Google Maps images paired with ground truth masks, this study evaluates models ranging from linear classifiers to advanced architectures like CNNs and U-Nets. By comparing these methods, it underscores the potential of machine learning to transform road detection and infrastructure planning.

II. MODELS AND METHODS

Road segmentation is framed as a pixel-wise classification task, where each pixel in a satellite image is classified as either road (denoted 1) or background (denoted as 0). Machine learning models process input features derived from the pixel values to estimate the likelihood of a pixel being part of a road. If the estimated probability meets or exceeds an initial threshold of 0.5, the pixel is classified as road; otherwise, it is classified as background. However, to ensure a better F1-score, a cross-validation is done at the end on the most promising models to determine the optimal threshold for classification.

A. Data set presentation

The given dataset consists of 100 google maps satellite colored images each measuring 400×400 pixels and represented as a 3-dimensional array ($400 \times 400 \times 3$) corresponding to the RGB color channels. These three channels (R, G, B) consist of the proportion of red, green and blue of each pixel. Paired with each image is a corresponding ground-truth mask in black and white, where white pixels represent roads, and black pixels denote the background.

B. Data Preprocessing

To improve model performance, an additional dataset was loaded into the training process. This supplementary dataset contains 1000 google maps satellite colored images of size $600 \times 600 \times 3$ and their corresponding pair of size $600 \times 600 \times 1$, which were respectively resized to $400 \times 400 \times 3$ and $400 \times 400 \times 1$ to align with the requirements of the architecture. This enhancement aimed to increase the diversity of training examples and improve segmentation accuracy.

C. Implementation

To achieve optimal accuracy, two approaches were explored. The first approach involves splitting the input images into 25×25 smaller patches of size 16×16 pixels. Each patch is classified as either road or background, and the classifications are then aggregated to reconstruct a 400×400 image, where all pixels in a given patch inherit the classification of the patch. This method employs classification algorithms such as Logistic Regression, Multi-Layer Perceptron (MLP), and Convolutional Neural Networks (CNNs).

The second approach directly predicts a $400 \times 400 \times 1$ binary mask from the $400 \times 400 \times 3$ input image. This is achieved using advanced neural architectures, including variations of MLPs, CNNs, and U-Nets, which process the entire image as a single unit, leveraging spatial relationships to enhance prediction accuracy.

D. Training

The training process was carefully designed to address the dataset's class imbalance, where background pixels significantly outnumbered road pixels. To tackle this, a weighted binary cross-entropy loss function was used, giving more importance to road pixels to ensure balanced predictions and improved segmentation accuracy. By applying a positive weight parameter given by $\frac{\#whites}{\#pixels}$, the loss function penalized misclassifications of road pixels more heavily, ensuring the model focused adequately on the minority class.

Optimization was handled using the AdamW optimizer, which effectively prevents overfitting while maintaining computational efficiency through adaptive learning rates and weight decay. Additionally, a dynamic learning rate scheduler (ReduceLROnPlateau) adjusted the learning rate whenever validation loss plateaued for three consecutive epochs, ensuring smooth and efficient convergence.

This standardized training setup, incorporating weighted loss, AdamW optimization, and a learning rate scheduler, was applied across all models, from logistic regression to U-Net.

It provided a robust foundation for handling the dataset's challenges and achieving reliable segmentation results.

E. patch splitting

The input images, sized $400 \times 400 \times 3$, are segmented into smaller patches of 25×25 , resulting in 625 sub-images per original image. Each patch represents an area of 16×16 pixels, amounting to 256 pixels per sub-image, as illustrated in Fig. 1. These sub-images are processed independently. For each corresponding ground truth patch, a classification label is assigned based on the proportion of white pixels: patches containing more than 25% 'road' pixels are labeled as road. This approach facilitates efficient computation and supports localized prediction capabilities.

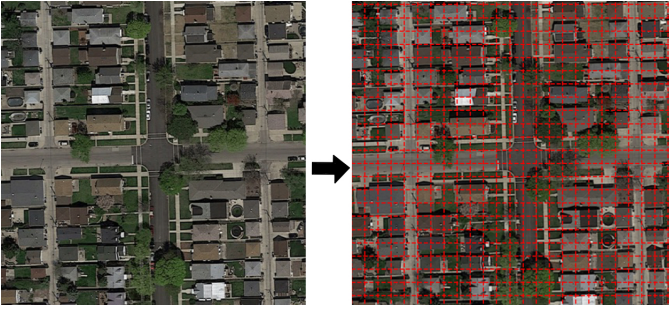


Fig. 1. Fraction of the input images into batches

1) *Logistic Regression*: Logistic regression, a widely used linear classification algorithm, predicts the probability of a given outcome, ranging between 0 and 1. The method employs two key hyperparameters: $\text{max-iter} = 100$, which defines the maximum number of iterations allowed for convergence, and $\gamma = 0.01$, which regulates the learning rate. In this implementation, logistic regression was applied independently to each of the 625 patches derived from the patch-splitting process, producing preliminary predictions for road segmentation on a localized basis.

2) *Multi-Layer Perceptron*: The Multi-Layer Perceptron (MLP) is a neural network that excels at identifying non-linear patterns in data. By utilizing layers of interconnected nodes, each applying mathematical transformations to its inputs, the MLP can model complex relationships that simpler methods, such as logistic regression, are unable to capture. In this study, the MLP was applied to 16×16 pixel patches created during the patch-splitting phase, with each patch processed independently. The network featured two hidden layers, each comprising 100 neurons, and used the ReLU (Rectified Linear Unit) activation function to introduce non-linearity. This enabled the MLP to learn intricate patterns within each patch. The model classified each patch as either road or background, contributing to the reconstruction of the segmented image.

3) *Convolution Neural Networks*: While the Multi-Layer Perceptron (MLP) significantly improved prediction accuracy compared to logistic regression by introducing non-linearity, further improvements were pursued by leveraging Convolutional Neural Networks (CNNs). CNNs are particularly well-suited for image data as they are designed to analyze spatial relationships between pixels within local regions of the input. CNNs utilize convolutional layers that apply filters over small patches of the image, enabling them to detect local patterns such as edges, textures, and shapes. For this implementation, the CNN was applied to each 16×16 pixel patch produced during patch splitting. The network featured four convolutional layers, progressively increasing the number of channels from 16 to 128 to capture increasingly complex features. These convolutional operations allowed the model to focus on the local spatial structure within each patch, significantly improving the accuracy of the predictions.

F. Holistic Processing

While patch classification provided encouraging results, it introduces limitations in precision due to the inherent grouping of pixels into 16×16 patches. When a patch contains both road and background pixels, the model assigns a single label to the entire patch, potentially leading to inaccuracies in the segmentation map. To overcome this issue, a holistic processing approach was implemented. This method directly reconstructs the entire output image from the input without splitting it into smaller patches. By treating the $400 \times 400 \times 3$ input image as a whole, this approach enables pixel-wise classification, capturing both local and global spatial information. This results in a more detailed and accurate segmentation map, eliminating the loss of precision caused by patch-level predictions.

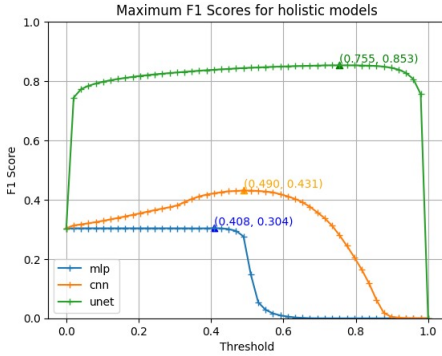
1) *Multi-Layer Perceptron*: In the holistic approach, the Multi-Layer Perceptron (MLP) processes the entire $400 \times 400 \times 3$ image directly, predicting a $400 \times 400 \times 1$ segmentation map. The architecture features three hidden layers with 500 neurons each and ReLU activations, enabling it to model complex, non-linear relationships. With 320,661,500 trainable parameters, the MLP flattens the input into a 1D vector for processing and reshapes the output back to its original dimensions for segmentation.

2) *Convolution Neural Networks*: Holistic Convolutional Neural Networks (CNNs) process entire $400 \times 400 \times 3$ images to produce $400 \times 400 \times 1$ segmentation maps, capturing both local and global spatial relationships. With 150,817 trainable parameters, the network uses four convolutional layers, pooling, and transposed convolutions for feature extraction and resolution adjustments. ReLU activations and a 0.3 dropout rate improve generalization, enabling smooth and accurate segmentation while addressing the limitations of patch-based methods.

3) *U-Nets*: The U-Net architecture enhances holistic road segmentation by combining spatial context with precise localization, making it ideal for tasks requiring accurate boundary detection. Composed of a contracting path (encoder) and an expanding path (decoder), U-Net downsamples the input image to extract high-level features and upsamples it to reconstruct a detailed segmentation map. Skip connections link encoder and decoder layers, preserving fine-grained spatial details and ensuring precise boundary delineation. The downsample and upsample lead to the use of 31043521 trainable parameters for this model.

Dropout, with a rate of 0.3, was incorporated into the training process to prevent overfitting by randomly deactivating neurons. This improvement encouraged the network to learn more generalized patterns, significantly increasing the F1-score and enhancing segmentation accuracy. By leveraging both local and global spatial information, U-Net effectively captures road continuity and delivers robust and precise results for road segmentation.

4) *Cross Validation*: Cross-validation was employed to determine the optimal threshold for classifying a pixel as road across all holistic models (MLP, CNN, and U-Net). The threshold was systematically varied, and its impact on the F1-score was evaluated on the validation set. Figure II-F4 illustrates the relationship between the threshold and the F1-score for each model, highlighting the threshold that maximized performance. For U-Net, the F1-score peaked at 0.853



with an optimal threshold of 0.755, showcasing its ability to balance precision and recall effectively. In comparison, the CNN and MLP achieved maximum F1-scores of 0.490 and 0.408 at thresholds 0.431 and 0.304, respectively. These results emphasize the importance of cross-validation in fine-tuning the classification process and demonstrate U-Net's superior performance in handling the complexities of road segmentation.

III. RESULTS

A. Patch splitting

Logistic regression encountered significant challenges in road segmentation, achieving an F1-score of 0.45 (Tab I), which fell short of producing reliable results. Its inability to

model non-linear relationships in the data made it unsuitable for the task and highlighted the need for more sophisticated methods. This limitation prompted the use of the Multi-Layer Perceptron (MLP), which introduced non-linearity and improved the F1-score to 0.51 (Tab I). The MLP was better at identifying complex patterns compared to logistic regression, marking a clear step forward in performance. However, processing each patch independently meant the MLP failed to consider spatial relationships between neighboring pixels, an essential feature for accurate road segmentation.

To address this, Convolutional Neural Networks (CNNs) were introduced, leveraging convolutional layers to capture both local patterns and spatial relationships within 16×16 patches. CNNs significantly outperformed both logistic regression and the MLP, achieving an F1-score of 0.70 (Tab I). The ability to distinguish road patterns from background noise by focusing on spatial structure was a key advantage of CNNs. When the patch-level predictions were aggregated to reconstruct the full 400×400 image, the results became more consistent and accurate, demonstrating the robustness of this approach.

These findings underscore the critical role of convolutional operations in extracting meaningful spatial features for road segmentation. CNNs proved to be a strong candidate for the task, effectively bridging the gap between local feature extraction and overall image reconstruction.

TABLE I
PERFORMANCE COMPARISON OF PATCH SPLITTING MODELS

Model	Accuracy	F1-Score
Logistic Regression	0.55	0.45
Multi-Layer Perceptron (MLP)	0.75	0.51
Convolutional Neural Networks (CNN)	0.83	0.70

B. Holistic Processing

The holistic Multi-Layer Perceptron (MLP) approach, which processed the entire image at once to avoid inconsistencies from patch-level aggregation, performed the worst among the tested models, achieving an F1-score of 0.14 (Tab II). This was primarily due to the loss of spatial context, as the MLP lacks the ability to recognize shapes or maintain a sense of pixel closeness. Reducing the 400×400×3 features to just 500 neurons further compounded this limitation, making the model unsuitable for road segmentation tasks.

Holistic Convolutional Neural Networks (CNNs), on the other hand, significantly outperformed the MLP by capturing both local and global spatial relationships, achieving an F1-score of 0.46 (Tab II). Unlike the MLP, the CNN preserved the notion of shape and structure in the image, thanks to its convolutional and pooling layers. By not excessively reducing the feature maps, the CNN retained important details like the outlines of houses or roads, allowing it to reconstruct more meaningful predictions. Despite higher computational demands, this balance of feature extraction and contextual understanding proved effective for road segmentation.

The U-Net architecture combined the strengths of both MLPs and CNNs, leveraging local detail from convolutional layers and maintaining global context through its encoder-decoder structure and skip connections. As a result, U-Net achieved the best performance, with an F1-score of 0.82 when trained on the primary dataset alone, an model prediction is shown in 5. However, its performance declined when trained on the combined dataset (Tab II), which included irrelevant content, such as lakes, confusing the model and reducing its accuracy. This underscores the importance of high-quality, relevant datasets for achieving optimal results, even with advanced models like U-Net.

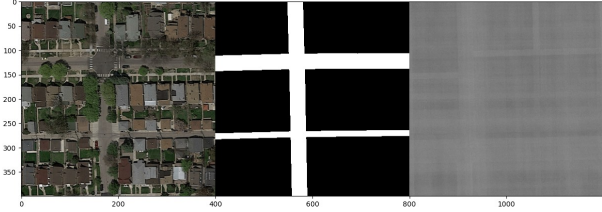


Fig. 2. Input, Expected Output and Prediction from trained MLP model

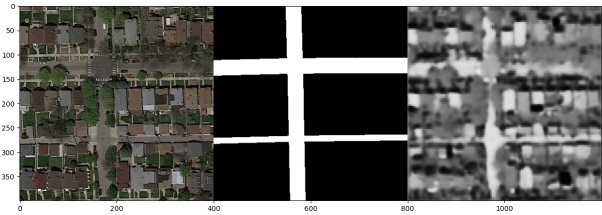


Fig. 3. Input, Expected Output and Prediction from trained CNN model

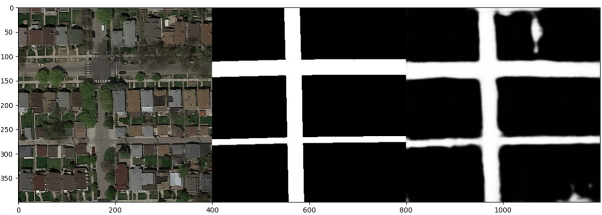


Fig. 4. Input, Expected Output and Prediction from trained U-net model

TABLE II
PERFORMANCE COMPARISON OF HOLISTIC MODELS

Model	Accuracy	F1-Score
Multi-Layer Perceptron (MLP)	0.69	0.31
Convolutional Neural Networks (CNN)	0.68	0.43
U-nets trained with satellite images	0.76	0.77
U-nets	0.92	0.82

IV. DISCUSSION

This study demonstrated a progression of machine learning methods for road segmentation, with U-Net achieving the best results. While training was conducted on GPUs for up to 50

epochs, the need to run the code on CPUs limited training to 30 epochs. This limitation was justified as the F1-score plateaued after 30 epochs as shown in Fig 5, highlighting the model's efficiency and early convergence.

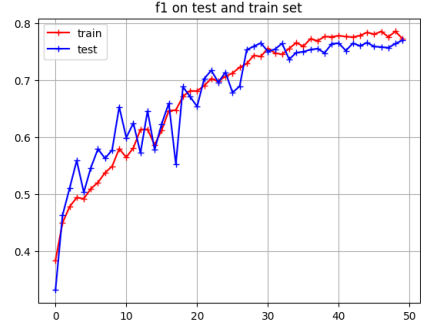


Fig. 5. Number of epochs

Future improvements could include optimizing U-Net for low-resource environments, employing transfer learning and attention mechanisms, and expanding datasets to enhance generalization. Additionally, post-processing techniques like Conditional Random Fields (CRFs) and real-time applications in autonomous navigation and disaster response could refine and validate these methods further.

V. CONCLUSION

This study explored machine learning methods for road segmentation, with U-Net standing out as the top performer, achieving an F1-score of 0.82. Its ability to combine local details and broader spatial context made it the most effective approach for creating accurate and reliable segmentation maps.

The results highlight the value of advanced architectures like U-Net for tackling complex pixel-level tasks, far surpassing simpler models. This demonstrates the potential of deep learning in applications such as urban planning, autonomous navigation, and disaster management.

Future work could focus on enhancing U-Net through techniques like transfer learning, attention mechanisms, or post-processing methods. Expanding and refining datasets and applying these models to real-world scenarios will help unlock their full potential for robust, automated road detection.

VI. ETHICS AND RISKS

Automated road segmentation systems present ethical challenges, particularly regarding data privacy, bias, and potential misuse. Satellite imagery used for training may inadvertently capture sensitive information, requiring strict compliance with privacy regulations and removal of identifiable features. Additionally, biased datasets can lead to unequal performance in underrepresented regions, emphasizing the importance of using diverse training data. There is also a risk of misuse, such as employing these systems for unauthorized surveillance, which necessitates the establishment of clear ethical guidelines and standards. By addressing these concerns, the technology can be deployed responsibly and equitably.